

SECTION 04810
UNIT MASONRY ASSEMBLIES

PART 1 GENERAL

1.1 SECTION INCLUDES

- A. Concrete Block.
- B. Clay Facing Brick.
- C. Mortar and Grout.
- D. Reinforcement and Anchorage.
- E. Flashings.
- F. Accessories.

1.2 REFERENCE STANDARDS

- A. ACI 530/530.1/ERTA - Building Code Requirements and Specification for Masonry Structures and Related Commentaries; American Concrete Institute International; 2011.
- B. ASTM A82/A82M - Standard Specification for Steel Wire, Plain, for Concrete Reinforcement; 2007.
- C. ASTM A153/A153M - Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware; 2009.
- D. ASTM A615/A615M - Standard Specification for Deformed and Plain Carbon Steel Bars for Concrete Reinforcement; 2014.
- E. ASTM A641/A641M - Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire; 2009a.
- F. ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2013.
- G. ASTM A1064/A1064M - Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete; 2013.
- H. ASTM A951 - Standard Specification for Masonry Joint Reinforcement.
- I. ASTM B209 - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate; 2010.
- J. ASTM B370 - Standard Specification for Copper Sheet and Strip for Building Construction; 2012.
- K. ASTM B695 - Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel.
- L. ASTM C55 - Standard Specification for Concrete Building Brick; 2011.
- M. ASTM C90 - Standard Specification for Loadbearing Concrete Masonry Units; 2014.

- N. ASTM C91/C91M - Standard Specification for Masonry Cement; 2012.
- O. ASTM C94/C94M - Standard Specification for Ready-Mixed Concrete; 2012.
- P. ASTM C129 - Standard Specification for Nonloadbearing Concrete Masonry Units; 2011.
- Q. ASTM C140/C140M - Standard Test Methods of Sampling and Testing Concrete Masonry Units and Related Units; 2014.
- R. ASTM C144 - Standard Specification for Aggregate for Masonry Mortar; 2011.
- S. ASTM C150/C150M - Standard Specification for Portland Cement; 2012.
- T. ASTM C207 - Standard Specification for Hydrated Lime for Masonry Purposes; 2006 (Reapproved 2011).
- U. ASTM C216 - Standard Specification for Facing Brick (Solid Masonry Units Made From Clay or Shale); 2014.
- V. ASTM C270 - Standard Specification for Mortar for Unit Masonry; 2012.
- W. ASTM C404 - Standard Specification for Aggregates for Masonry Grout; 2011.
- X. ASTM C476 - Standard Specification for Grout for Masonry; 2010.
- Y. ASTM C780 - Standard Test Method for Preconstruction and Construction Evaluation of Mortars for Plain and Reinforced Unit Masonry; 2012.
- Z. ASTM C1019 - Standard Test Method for Sampling and Testing Grout.
- AA. ASTM C1142 - Standard Specification for Extended Life Mortar for Unit Masonry.
- BB. ASTM C1357 - Standard Test Methods for Evaluating Masonry Bond Strength; 2009.
- CC. SMACNA (ASMM) - Architectural Sheet Metal Manual; Sheet Metal and Air Conditioning Contractors' National Association; 2012.
- DD. TMS MSJC - Building Code for Masonry Structures (ACI 530/ASCE 5/TMS 402), Specification for Masonry Structures (ACI 530.1/ASCE 6/TMS 602) and Commentaries.

1.3 SUBMITTALS

- A. See Section 01 33 00 - Submittals, for submittal procedures.
- B. Product Data: Provide data for masonry units, fabricated wire reinforcement, mortar, and masonry accessories.
- C. Manufacturer's Certificate: Certify that masonry units meet or exceed specified requirements.
- D. Manufacturer's Certificate: Certify that water repellent admixture manufacturer has certified masonry unit manufacturer as an approved user of water repellent admixture in the manufacture of concrete block.
- E. Buy American Act Certification: Submit documentation certifying that products comply with provisions of the Buy American Act BAA - 41 U.S.C. §§ 10a-10d

1.4 PRE-INSTALLATION MEETING

- A. Convene minimum one week prior to commencing work of this section. Notify Contracting Officer seven days in advance.
- B. Coordinate installation of all trades that penetrate the veneer system.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, handle, and store masonry units by means that will prevent mechanical damage and contamination by other materials.
- B. Stack material to prevent twisting, bending, and abrasion, and to provide ventilation. Slope metal sheets to ensure drainage.
- C. Prevent contact with materials that could cause discoloration or staining.

PART 2 PRODUCTS

2.1 CONCRETE MASONRY UNITS

- A. Concrete Block: Comply with referenced standards and as follows:
 - 1. Concrete Masonry Units: Concrete Masonry Unit Size and Shape: Nominal modular size of 8 x 8 x 16 inches and other sizes as indicated on drawings. Furnish special units for 90 degree corners, bond beams, lintels and pilasters.
 - 2. Load Bearing Concrete Masonry Units (CMU): Provide both hollow and solid units per ASTM C90; lightweight, with maximum 40% total water absorption.
 - 3. Aggregate for concrete masonry units shall conform to ASTM C331. Bottom ash aggregates are not permitted.
 - 4. The color of the mortar to be standard gray unless noted otherwise.

2.2 BRICK UNITS

- A. Facing Brick: ASTM C216, Type FBS, Grade SW.
 - 1. Color and texture: To match existing brick on 241st Operations Building and to match approved sample.
 - 2. Nominal size: As indicated on drawings and to match existing brick.
 - 3. Special shapes: Molded units as required by conditions indicated, unless standard units can be sawn to produce equivalent effect.

2.3 MORTAR AND GROUT MATERIALS

- A. Masonry Cement: ASTM C91/C91M, Type N.
 - 1. Colored mortar: Premixed cement as required to match TEC Campus Buildings.
- B. Portland Cement: ASTM C150/C150M, Type I; color as required to produce approved color sample.
- C. Hydrated Lime: ASTM C207, Type S.
- D. Mortar Aggregate: ASTM C144.
- E. Grout Aggregate: ASTM C404.
- F. Water: Clean and potable.

- G. Bonding Agent: Latex type.
- H. Mortar Mixing
 - 1. Thoroughly mix mortar ingredients using mechanical batch mixer, in accordance with ASTM C270 and in quantities needed for immediate use.
 - 2. Maintain sand uniformly damp immediately before the mixing process.
 - 3. Do not use anti-freeze compounds to lower the freezing point of mortar.
 - 4. If water is lost by evaporation, re-temper only within two hours of mixing.
- I. Grout Mixing
 - 1. Mix grout in accordance with ASTM C94/C94M.
 - 2. Thoroughly mix grout ingredients in quantities needed for immediate use in accordance with ASTM C476 for fine and coarse grout.

2.4 REINFORCEMENT AND ANCHORAGE

- A. Manufacturers of Joint Reinforcement and Anchors:
 - 1. Hohmann & Barnard, Inc: www.h-b.com.
 - 2. Substitutions: See Section 01600 - Materials, Equipment, and Substitutions.
- B. Joint Reinforcement: Use ladder type joint reinforcement where vertical reinforcement is involved and truss type elsewhere, unless otherwise indicated.
- C. Single Wythe Joint Reinforcement: Truss or ladder type; ASTM A1064/A1064M steel wire, mill galvanized to ASTM A641/A641M, Class 3; 0.1483 inch side rods with 0.1483 inch cross rods; width as required to provide not more than 1 inch and not less than 1/2 inch of mortar coverage on each exposure.
- D. Flexible Anchors: 2-piece anchors that permit differential movement between masonry and building frame, sized to provide not more than 1 inch and not less than 1/2 inch of mortar coverage from masonry face.
- E. Masonry Veneer Anchors: 2-piece anchors that permit differential movement between masonry veneer and structural backup, hot dip galvanized to ASTM A 153/A 153M, Class B.
 - 1. Anchor plates: Not less than 0.075 inch thick, designed for fastening to structural backup through sheathing by two fasteners.
 - 2. Anchor design to accommodate rigid insulation with wire legs firmly holding insulation in contact with structural backup.
 - 3. Wire ties: Manufacturer's standard shape, 0.1875 inch thick.
 - 4. Vertical adjustment: Not less than 3-1/2 inches.
 - 5. Expansion Pins: For attachment of masonry anchors to existing CMU.

2.5 FLASHINGS

- A. Pre-Finished Aluminum: ASTM B209; 0.032 inch thick; plain finish shop pre-coated with modified silicone coating.
 - 1. Modified silicone polyester coating: pigmented organic coating system, AAMA 2603; bakes enamel finish system.
- B. Through Wall Flashing: Copper Tuft, manufactured by Hohmann & Barnard.
 - 1. Polymer coating bonded to 2 oz. copper sheet.
 - 2. Fiberglass scrim embedded between coating and copper.
 - 3. Puncture and tear resistant.
 - 4. Asphalt free.
 - 5. Inside and Outside corner flashing, and end dams at window sills.

- C. Fasteners: Galvanized steel, with soft neoprene washers.
- D. Primer: Zinc chromate type.
- E. Lap Sealant: Type A.
- F. Plastic Cement: ASTM D4586, Type I.
- G. Control Joints: #RS-Standard, manufactured by Hohmann & Barnard. Extruded Rubber. ASTM D2000 2AA-805. Length to match masonry dimension.
- H. Fabrication
 - 1. Form sections true to shape, accurate in size, square, and free from distortion or defects.
 - 2. Form pieces in longest possible lengths.
 - 3. Hem exposed edges on underside 1/2 inch; miter and seam corners.
 - 4. Fabricate corners from one piece with minimum 18 inch long legs; seam for rigidity, seal with sealant.
 - 5. Form material with flat lock seams, except where otherwise indicated. At moving joints, use sealed lapped, bayonet-type or interlocking hooked seams.

2.6 ACCESSORIES

- A. Weeps: Hohmann & Barnard, Inc; Product: QV Quandro - vent at top and bottom (where the thru the wall flashing occurs) of clay masonry veneer. Standard size brick. Color: clear.
- B. Cleaning Solution: Non-acidic, not harmful to masonry work or adjacent materials.

2.7 MORTAR AND GROUT MIXES

- A. Mortar for Unit Masonry: ASTM C270, using the Proportion Specification.

PART 3 EXECUTION

3.1 EXAMINATION

- A. Verify that field conditions are acceptable and are ready to receive masonry.
- B. Verify that related items provided under other sections are properly sized and located.
- C. Verify that built-in items are in proper location, and ready for roughing into masonry work.

3.2 PREPARATION

- A. Direct and coordinate placement of metal anchors supplied for installation under other sections.
- B. Provide temporary bracing during installation of masonry work. Maintain in place until building structure provides permanent bracing.

3.3 COLD AND HOT WEATHER REQUIREMENTS

- A. Comply with requirements of ACI 530/530.1/ERTA or applicable building code, whichever is more stringent.

3.4 COURSING

- A. Establish lines, levels, and coursing indicated. Protect from displacement.
- B. Maintain masonry courses to uniform dimension. Form vertical and horizontal joints of uniform thickness.
- C. Concrete Masonry Units:
 - 1. Bond: Running.
 - 2. Coursing: One unit and one mortar joint to equal 8 inches.
 - 3. Mortar Joints: Concave.
- D. Brick Units:
 - 1. Bond: To match existing.
 - 2. Coursing: Match existing brick coursing.
 - 3. Mortar Joints: Concave to match existing.

3.5 PLACING AND BONDING

- A. Lay solid masonry units in full bed of mortar, with full head joints, uniformly jointed with other work.
- B. Lay hollow masonry units with face shell bedding on head and bed joints.
- C. Buttering corners of joints or excessive furrowing of mortar joints is not permitted.
- D. Remove excess mortar and mortar smears as work progresses.
- E. Remove excess mortar with water repellent admixture promptly. Do not use acids, sandblasting or high pressure cleaning methods.
- F. Interlock intersections and external corners, except for units laid in stack bond.
- G. Do not shift or tap masonry units after mortar has achieved initial set. Where adjustment must be made, remove mortar and replace.
- H. Perform job site cutting of masonry units with proper tools to provide straight, clean, unchipped edges. Prevent broken masonry unit corners or edges.
- I. Cut mortar joints flush where wall tile is scheduled or resilient base is scheduled.

3.6 WEEPS/CAVITY VENTS

- A. Install weeps in veneer and cavity walls at 32 inches on center horizontally above through-wall flashing, above shelf angles and lintels, and at bottom of walls.

3.7 CAVITY MORTAR CONTROL

- A. Do not permit mortar to drop or accumulate into cavity air space or to plug weep/cavity vents.
- B. For cavity walls, build inner wythe ahead of outer wythe to accommodate accessories.
- C. Install cavity mortar control panels continuously throughout full height of exterior masonry cavities during construction of exterior wythe, complying with manufacturer's installation instructions. Verify that airspace width is no more than 3/8 inch greater than panel thickness. Install horizontally between joint reinforcement. Stagger end joints in adjacent rows. Fit to perimeter construction and penetrations without voids.

- D. Install cavity mortar diverter at base of cavity and at other flashing locations as recommended by manufacturer to prevent mortar droppings from blocking weep/cavity vents.

3.8 REINFORCEMENT AND ANCHORAGE – GENERAL

- A. Unless otherwise indicated on drawings or specified under specific wall type, install horizontal joint reinforcement 16 inches on center.
- B. Place masonry joint reinforcement in first and second horizontal joints above and below openings. Extend minimum 16 inches each side of opening.
- C. Place continuous joint reinforcement in first and second joint below top of walls.
- D. Lap joint reinforcement ends minimum 6 inches.
- E. Reinforce stack bonded unit joint corners and intersections with strap anchors 16 inches on center.
- F. Fasten anchors to structural framing and embed in masonry joints as masonry is laid. Unless otherwise indicated on drawings or closer spacing is indicated under specific wall type, space anchors at maximum of 36 inches horizontally and 24 inches vertically.

3.9 REINFORCEMENT AND ANCHORAGE - SINGLE WYTHE MASONRY

- A. Install horizontal joint reinforcement 8 inches on center.
- B. Place masonry joint reinforcement in first and second horizontal joints above and below openings. Extend minimum 16 inches each side of opening.
- C. Place continuous joint reinforcement in first and second joint below top of walls.
- D. Lap joint reinforcement ends minimum 6 inches.

3.10 REINFORCEMENT AND ANCHORAGE - MASONRY VENEER

- A. Install horizontal joint reinforcement 16 inches on center.
- B. Place masonry joint reinforcement in first and second horizontal joints above and below openings. Extend minimum 16 inches each side of opening.
- C. Place continuous joint reinforcement in first and second joint below top of walls.
- D. Lap joint reinforcement ends minimum 6 inches.
- E. Masonry Back-Up: Embed anchors to bond veneer at maximum 16 inches on center vertically and 36 inches on center horizontally. Place additional anchors at perimeter of openings and ends of panels, so maximum spacing of anchors is 8 inches on center.

3.11 MASONRY FLASHINGS

- A. Whether or not specifically indicated, install masonry flashing to divert water to exterior at all locations where downward flow of water will be interrupted.
 - 1. Extend flashings full width at such interruptions and at least 6 inches into adjacent masonry or turn up at least 8 inches to form watertight pan at non-masonry construction.
 - 2. Remove or cover protrusions or sharp edges that could puncture flashings.
 - 3. Seal lapped ends and penetrations of flashing before covering with mortar.

3.12 CLEANING

- A. Remove excess mortar and mortar droppings.
- B. Replace defective mortar. Match adjacent work.
- C. Clean soiled surfaces with cleaning solution.
- D. Use non-metallic tools in cleaning operations.

3.13 PROTECTION

- A. Without damaging completed work, provide protective boards at exposed external corners that are subject to damage by construction activities.

END OF SECTION